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Effects of the Nordic Hamstring exercise on sprint capacity in male football players: a randomized controlled trial

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ABSTRACT

This assessor-blinded, randomized controlled superiority trial investigated the efficacy of the 10-week Nordic Hamstring exercise (NHE) protocol on sprint performance in football players.

Thirty-five amateur male players (age: 17–26 years) were randomized to a do-as-usual control group (CG; $n = 17$) or to 10-weeks of supervised strength training using the NHE in-season (IG; $n = 18$). A repeated-sprint test, consisting of 4×6 10 m sprints, with 15 s recovery period between sprints and 180 s between sets, was conducted to evaluate total sprint time as the primary outcome. Secondary outcomes were best 10 m sprint time (10mST) and sprint time during the last sprint (L10mST). Additionally, peak eccentric hamstring strength (ECC- P_{HS}) and eccentric hamstring strength capacity (ECC-CAP $_{HS}$) were measured during the NHE.

Ten players were lost to follow-up, thus 25 players were analyzed (CG $n = 14$; IG $n = 11$). Between-group differences in mean changes were observed in favor of the IG for sprint performance outcomes; TST (-0.649 s, $p = 0.056$, $d = 0.38$), 10mST (-0.047 s, $p = 0.005$, $d = 0.64$) and L10mST (-0.052 s, $p = 0.094$, $d = 0.59$), and for strength outcomes; ECC- P_{HS} (62.3 N, $p = 0.006$, $d = 0.92$), and ECC-CAP $_{HS}$ (951 N, $p = 0.005$, $d = 0.95$).

In conclusion, the NHE showed small-to-medium improvements in sprint performance and large increases in peak eccentric hamstring strength and capacity.

Trial Registration Number: NCT02674919

ARTICLE HISTORY

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KEYWORDS

Resistance training; hamstring muscles; repeated-sprint ability; high-speed running; injury prevention

Introduction

Maximal sprinting and acceleration activities represent <5% of the total distance covered during professional football matches, but are essential physical determinants of match performance (Barnes, Archer, Hogg, Bush, & Bradley, 2014). Sprinting is also the predominant injury situation in hamstring strain injuries (HSIs) (Askling, Tengvar, Saartok, & Thorstensson, 2007), which are the most frequent type of muscle injury in football (Ekstrand, Hagglund, & Walden, 2011). Risk factors for HSIs include low levels of maximal eccentric hamstring strength measured during the Nordic hamstring exercise (NHE), and short biceps femoris long head (BFLh) fascicle length (Timmins et al., 2016). Additionally, reduced endurance capacity of the hamstring muscles has been proposed as a potential risk factor (Schuermans, Van Tiggelen, Danneels, & Witvrouw, 2016).

The NHE is an on-field partner-assisted eccentric hamstring exercise, which increases eccentric hamstring strength (Mjolsnes, Arnason, Osthaugen, Raastad, & Bahr, 2004) and BFLh fascicle lengths (Bourne et al., 2017). Consequently, the progressive 10-week NHE protocol introduced by Mjolsnes et al. (2004) effectively reduces the incidence of HSIs in football (Petersen, Thorborg, Nielsen, Budtz-Jorgensen, & Holmich, 2011). Despite the high

quality evidence of the effectiveness of the NHE to reduce HSIs, implementation of the exercise in elite football remains limited (Bahr, Thorborg, & Ekstrand, 2015). This could be due to sports medicine and coaching professionals labelling the NHE as ineffective (Ekstrand, Hagglund, Kristenson, Magnusson, & Walden, 2013) or even “contraindicated” (Gambetta & Benton, 2006) in relation to injury prevention and performance, respectively.

Theoretically, increased eccentric hamstring strength and neuromuscular activity could improve sprint performance. In sprinting mechanics, the hamstring muscles decelerate knee extension in the swing phase, and subsequently generate horizontal forces in the stance phase through hip extension (Morin et al., 2015). Gains in eccentric hamstring strength induced by a variety of exercise modalities have previously shown promising effects on sprint performance in football players (Askling, Karlsson, & Thorstensson, 2003; Mendiguchia et al., 2015). However, studies investigating the effect of the NHE on sprint performance are lacking. To date, only a single pilot study from our group has investigated sprint performance using the Mjolsnes 10-week injury-prevention NHE protocol (Mjolsnes et al., 2004), and reported improved 10 m sprint performance (Krommes et al., 2017).

Therefore, this trial aimed to investigate the efficacy of a 10-week in-season NHE training intervention on sprint performance in male football players. It was hypothesized that NHE training would improve sprint performance compared to control subjects performing football training as usual.

Methods

Trial design

This is the primary trial report of a single assessor-blinded randomized controlled superiority trial with a 2-group parallel design. The intervention was a supervised 10-week NHE protocol (Mjolsnes et al., 2004) and the comparator was a do-as-usual group. The primary endpoint was repeated-sprint performance recorded at 10-week follow-up (immediately after the intervention). The study protocol was registered at clinicaltrials.gov prior to inclusion of the first participant (ID: NCT02674919) on the 2nd of February 2016. The reporting adheres to the Consolidated

Standards of Reporting Trials (CONSORT) (Moher et al., 2010) and Ethical approval was obtained from the Danish Ethics Committee of the Capital Region (ID: H-15020709).

Participants

Thirty-five amateur male football players (4th tier) in the Greater Copenhagen area were included through convenience sampling based on geographic location of football clubs (Figure 1). The inclusion criteria were: Male football players participating in full team training at age of 17–35 years. Exclusion criteria were: any specific systematic eccentric strength training of the hamstring muscles or any specific systematic sprint performance training >1 session/week in the preceding 2 months prior to study initiation; hamstring muscle pain resulting in absence from football play in the preceding 2 months prior to study initiation; any longstanding injury (>6 weeks) in the lower extremities during the preceding 6 months prior to study initiation. Furthermore, if players

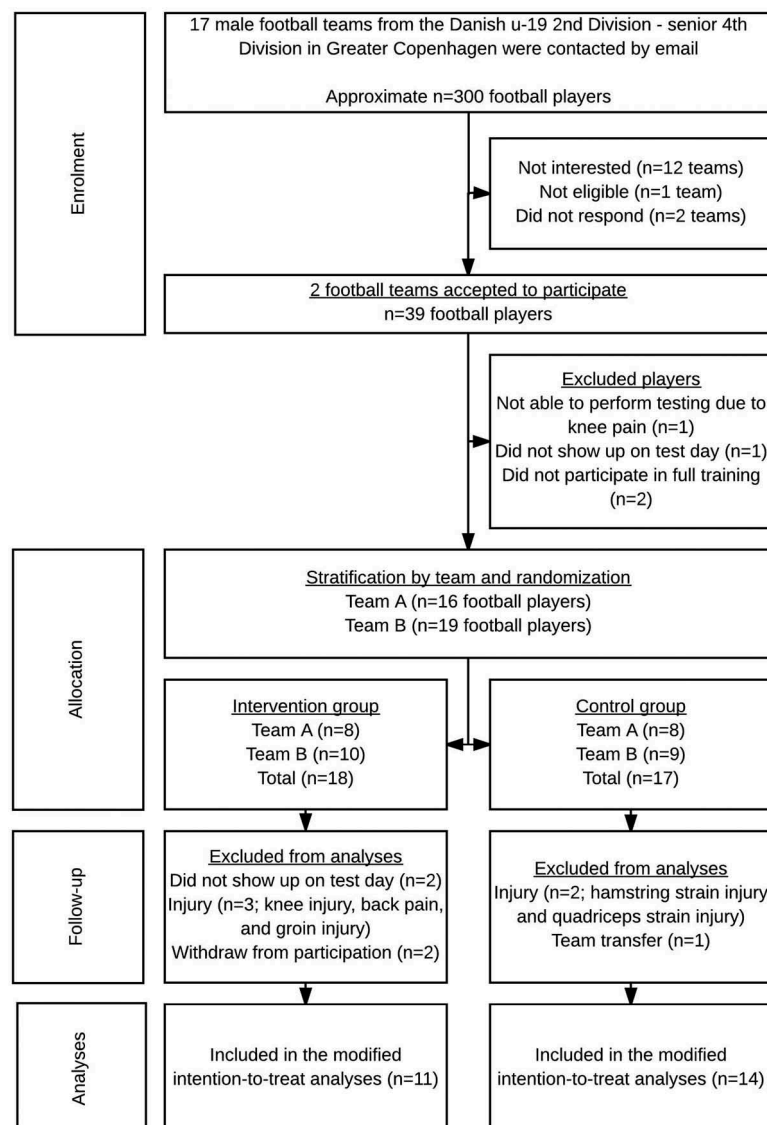


Figure 1. Flow diagram of participant enrolment, allocation, follow-up, and analyses.

experienced pain in the hamstring muscles during the NHE or due to injury, this was reported to the physiotherapist who supervised the training intervention. Subsequently, players were excluded by a blinded assessor based on “pain in hamstring muscles”. All participants gave written informed consent according to the Declaration of Helsinki.

Study settings

The study was conducted during the second and last part of the season period (March to June 2016), when the players typically had three football training sessions (approx. 5–7 hours of training) and one match per week. All experimental data were collected using the same indoor arena with artificial turf, located in the Greater Copenhagen area. The participants were instructed to refrain from vigorous physical activity 24 h prior to testing.

Intervention

The intervention is reported according to the Template for Intervention Description and Replication (TIDieR) checklist (Hoffmann et al., 2014), and described in detail using the mechano-biological exercise descriptors of resistance exercise stimuli (supplementary file) (Toigo & Boutellier, 2006). The intervention group performed a 10-week progressive eccentric strengthening program for the hamstring muscles using the NHE (Table 1) (Mjolsnes et al., 2004). The NHE is a partner-assisted exercise, performed without additional equipment or external load. In a kneeling position (90° of knee flexion), a partner fixates the ankles to ensure constant contact between the lower legs and the ground throughout the movement. Subsequently, the person leans forward as slowly as possible, while attempting to resist the forward motion by maximally activating the hamstring muscles in the eccentric phase. Participants were asked to maintain a straight torso and slightly flexed hips during the whole range of motion. Participants were asked to use their hands to buffer the fall after they “let go” and to forcefully push themselves back to the starting position with their hands to minimize hamstring force production in the corresponding concentric contraction phase (Mjolsnes et al., 2004). Players were instructed to alternate between performing one set and manually fixating the ankles of the partner player, respectively. Recovery time in-between exercise sessions were 48h. All NHE training was performed as group training sessions subsequent to regular football training sessions to avoid that exercise-induced hamstring muscle fatigue would interfere with the regular football training (Marshall, Lovell, Knox, Brennan, & Siegler, 2015). Sessions were supervised by a trained physiotherapist, who also registered training compliance of the NHE protocol as

completed number of sessions. To minimize the risk of between-group contamination the NHE was only introduced to the intervention group. The control group continued to play football as usual. Both groups were instructed not to change training regime during the intervention period. This included avoiding addition of specific sprint training or specific eccentric hamstring strength training.

Outcomes

The primary outcome measure was total sprint time (TST) in seconds, calculated as the sum of 4×6 maximal 10 m sprints recorded during a repeated-sprint test. The TST was chosen as primary outcome measure based on its high reliability (ICC: 0.91; % TE: 1.5) (Gabbett, 2010). Single sprint times were measured using dual-beam photoelectric cells (Witty Wireless Training Timer, Microgate, Bolzano, Italy). Each 10 m sprint was performed from a standing position with the front foot placed 1 m behind the photocell. Each sprint was initiated every 15 s and every 6 sprints (set) were separated by 3 min of active rest. Typically, repeated-sprint tests use a sprint distance of ≥ 30 m (Buchheit, Mendez-Villanueva, Delhomel, Brughelli, & Ahmaidi, 2010; Campos-Vazquez et al., 2015), however, the validity of such distance in relation to football match play has been questioned (Schimpchen, Skorski, Nopp, & Meyer, 2016). Therefore, it has been proposed to use shorter sprint distances (Taylor, Macpherson, Spears, & Weston, 2016) to better replicate the physical characteristics (multiple accelerations) of modern football match play (Barnes et al., 2014).

For the secondary outcome measures, fastest 10 m sprint (10mST) and sprint time of the last 10 m sprint (L10mST) were obtained during the repeated-sprint test. Additionally, peak eccentric hamstring strength (ECC- P_{HS}) and hamstring endurance capacity were evaluated using a prototype custom-made testing device designed to measure eccentric knee flexor strength during the NHE (Opar, Piatkowski, Williams, & Shield, 2013). Participants were kneeling on a padded board with ankles secured superior to the malleoli by individual braces attached to wireless load cells with a 100 Hz A/D sampling rate (Delphi Force Measurement, Gold Coast, Australia). From this position, participants performed the NHE. Three warm-up repetitions followed by 15 repetitions were performed. A trial was deemed acceptable when the force output reached a peak force value, followed by a rapid decline in force. ECC- P_{HS} was calculated as the mean peak force of both legs during the initial three repetitions (Opar et al., 2013). Eccentric hamstring endurance was evaluated as strength fatigue resistance (SFR $_{HS}$) and eccentric hamstring strength capacity (ECC-CAP $_{HS}$). SFR $_{HS}$ was defined as the mean rate of decline in peak force per repetition from repetition 1 to 15. ECC-CAP $_{HS}$ was defined as the total sum of peak force recorded in each of the repetitions performed. Finally, hamstring muscle flexibility was measured as range of motion of active knee extension (AKE) (Reurink et al., 2013). The test followed a standardized procedure (Reurink et al., 2013) using a two-part digital inclinometer (DUALER™ IQ, Jtech Medical, Salt Lake City, Utah, USA). In supine position, the participants were asked to extend the knee while maintaining 90° of hip flexion. AKE was defined as the angle between tibia and the horizontal plane and reported as mean of both legs (Reurink et al., 2013).

Table 1. Nordic Hamstring exercise intervention protocol.

Week	Sessions/week	Sets	Repetitions
1	1	2	5
2	2	2	6
3	3	3	6–8
4	3	3	8–10
5–10	3	3	12, 10, 8

Prior to testing participants underwent a standardized 15 min warm-up procedure consisting of running drills and sprinting with increasing intensity. The repeated-sprint and hamstring strength tests were separated by 15 min of rest.

Sample size

An a-priori sample size calculation based on the primary outcome measure, TST, was performed using G*power software (v.3.1, Heinrich-Heine-Universität, Düsseldorf, Germany). The expected mean was set at 45 s with a hypothesized improvement of 2.25 s, equivalent to 5%, and a standard deviation (SD) of 2 s (Gabbett, 2010). Accepting 80% statistical power and 5% type-1 error risk, 28 participants were deemed to be adequate for detecting a large effect size ($d \geq 0.8$) (Cohen, 1992). Due to convenience and to account for potential drop-outs during the intervention period, we aimed at including all players from two teams.

Randomization

Following baseline testing, a person not involved in the study conducted the randomization. Participants were stratified by club, and individually randomized to the intervention- or control group (allocation ratio 1:1) using sealed, opaque envelopes each containing the name of a single player. Players were alternately allocated to a group starting with the intervention group. The pre-determined allocation sequence was concealed from the person involved in the randomization. Randomization was conducted for one team at a time.

Assessor blinding

One tester (LI), who was blinded to group allocation, conducted all test activities at baseline (immediately before the intervention) and at follow-up (immediately after the intervention).

Statistical methods

For the analysis of the primary (TST) and secondary (10mST, L10mST, and AKE) outcomes, independent student t-tests were employed to analyze between-group differences in mean pre-post changes. Due to apparent between-group baseline differences in ECC-P_{HS} and ECC-CAP_{HS} highlighted during the review process we adopted an Analysis of Covariance (ANCOVA) using the baseline score as the covariate (Vickers & Altman, 2001) instead of the a-priori planned independent student t-tests. Paired student t-tests were used to evaluate within-group differences for TST, 10mST, L10mST, ECC-P_{HS}, ECC-CAP_{HS}, and AKE. SFR_{HS} was analyzed using mixed-effect regression models with random intercept and slope to test difference in rate of force decline between and within groups. Data analysis was performed in SPSS (v. 23, IBM Corporation, New York, USA). The dependent variables showed normal distribution (Shapiro–Wilk test and q-q plots). A modified intention-to-treat (mITT) approach including all participants with baseline and follow-up measures, and a per-protocol approach including participants with a compliance rate $\geq 70\%$ were planned for all analyses. However, as only few players had a compliance rate

of $\geq 70\%$, all analyses were based on a mITT approach. A mITT analysis was used over an ITT analysis as the present study is the first to investigate the efficacy of the NHE on repeated-sprint performance. Therefore, we were interested in the efficacy of the NHE given that a minimal dose of training was performed, and thus only players with baseline and follow-up measures were included. For practical application the effect of the intervention was assessed using magnitude-based inferences (Hopkins, Marshall, Batterham, & Hanin, 2009) for TST, 10mST, L10mST, ECC-P_{HS}, ECC-CAP_{HS}, and AKE using an excel spreadsheet (Hopkins, 2003). Between-group Cohen's d was calculated using pre-intervention pooled SD and assessed as trivial ($d < 0.2$), small ($d \geq 0.2$), medium ($d \geq 0.5$), and large ($d \geq 0.8$) (Cohen, 1992). The smallest worthwhile difference (SWD) was set at 0.2 between-subject SD, suggesting to represent a clinically relevant change in performance in team sport athletes (Hopkins et al., 2009). Probabilities of lower, similar, or higher differences than the SWD were evaluated qualitatively using the following scale: $<0.5\%$, *almost certainly not*; ≥ 0.5 – 5% , *very unlikely*; >5 – 25% , *unlikely*; >25 – 75% , *possibly*; >75 – 95% , *likely*; >95 – 99.5% , *very likely*; $>99.5\%$, *almost certainly* (Hopkins et al., 2009). If the chance of both lower and higher values was $\geq 5\%$, the effect was defined as unclear (Hopkins et al., 2009). Data are presented as mean values (SD), unless otherwise stated. Qualitative inferences are presented with the corresponding chances of a negative, trivial, or positive effect, e.g. (0/17/83).

Results

Participants

Eligible participants were included from the 15th–24th of March 2016. Ten participants were lost to follow-up due to injuries unrelated to the NHE, withdrawal from study participation, team transfer, and non-attendance due to illness on test days. Thus 25 participants (intervention group $n = 11$, control group $n = 14$) were included in the analyses of TST, 10mST, L10mST, and AKE (Figure 1). Table 2 shows demographic baseline data of participants. Due to technical errors related to the hamstring testing device resulting in missing data, 20 participants (intervention group $n = 10$, control group $n = 10$) were included in the analysis of ECC-P_{HS}, whereas 19 participants (intervention group $n = 9$, control group $n = 10$) were included in the analysis of SFR_{HS} and ECC-CAP_{HS}. The intervention group had a mean compliance of 60.2% (12.2).

Primary outcome measure

A non-significant, but *likely* (0/17/83), small between-group difference in mean change in TST (-0.798 s vs. -0.149 s; mean difference, -0.649 s, $p = 0.056$, $d = 0.38$) was observed in favor of the intervention group (Table 3, Figure 2(a,b)).

Table 2. Demographic data of included participants at baseline showed as mean (SD).

	Intervention ($n = 11$)	Control ($n = 14$)
Age (years)	19.1 (1.8)	19.4 (2.1)
Stature (cm)	180.7 (7.3)	180.7 (6.5)
Body mass (kg)	76.2 (11.9)	77.0 (8.7)

Table 3. Overview of primary and secondary outcome measures in the intervention and control group. Data presented as mean (SD) unless stated otherwise.

	Intervention (n = 11)			Control (n = 14)			Between-group differences in mean change [95% CI], Cohen's d^a &
	Pre	Post	Within-group changes [95% CI]	Pre	Post	Within-group changes [95% CI]	
TST (s)	43.295 (2.81)	42.497 (2.26)*	-0.798 [-1.4;-0.2]	43.707 (1.26)	43.557 (1.24)	-0.149 [-0.6;0.3]	-0.649 [-1.318; 0.020], 0.38
10mST (s)	1.703 (0.10)	1.659 (0.08)*	-0.045 [-0.07;-0.02]	1.723 (0.05)	1.725 (0.05)	0.002 [-0.02;0.02]	-0.047 [-0.077; -0.016], 0.64**
L10mST (s)	1.866 (0.11)	1.806 (0.10)*	-0.06 [-0.12;-0.00]	1.854 (0.07)	1.846 (0.06)	-0.008 [-0.04;0.02]	-0.052 [-0.114; 0.010], 0.59
ECC-P _{HS} (N) $\wedge$	321.5 (50.7)	383.2 (68.6)*	61.7 [27.7;95.7]	382.0 (70.9)	372.7 (62.1)	-9.3 [-30.6;12.0]	62.3 [20.0; 104.5], 0.92**
SFR _{HS} (N/rep) $\wedge$	-3.3 (2.97)	-4.48 (3.87)	-1.2 [-4.4;2.1]	-2.0 (4.21)	-3.5 (3.73)	-1.5 [-4.3;1.3]	-1.0 [-3.2; 1.2]
ECC-CAP _{HS} (N) $\wedge$	4433 (653)	5203 (966)*	769 [281;1258]	5485 (1022)	5197 (918)	-288 [-591;15]	951 [323; 1579], 0.95**
AKE (degrees)	67.0 (13.3)	69.2 (13.8)	2.2 [-4.4;8.8]	73.3 (9.5)	74.7 (9.2)	1.4 [-1.1;3.9]	0.8 [-6.1; 7.6], 0.07

TST (total sprint time); 10mST (fastest 10 meter sprint time); L10mST (sprint time of last sprint trial); ECC-P_{HS} (peak eccentric hamstring muscle strength); SFR_{HS} (hamstring strength fatigue resistance); ECC-CAP_{HS} (eccentric hamstring strength capacity); AKE (active knee extension); s (seconds); N (Newton); N/rep (Newton/repetition). * Pre-to-post changes ($p < 0.05$). ** Between-group difference in mean change ($p < 0.01$). $\wedge$ Due to technical errors during testing 20 participants (intervention $n = 10$, control $n = 10$) are included in ECC-P_{HS} and 19 participants (intervention $n = 9$, control $n = 10$) are included in SFR_{HS} and ECC-CAP_{HS}. ^a Adjusted between-group differences in mean change for ECC-P_{HS} and ECC-CAP_{HS}. & Between-group difference at follow-up displayed for SFR_{HS}.

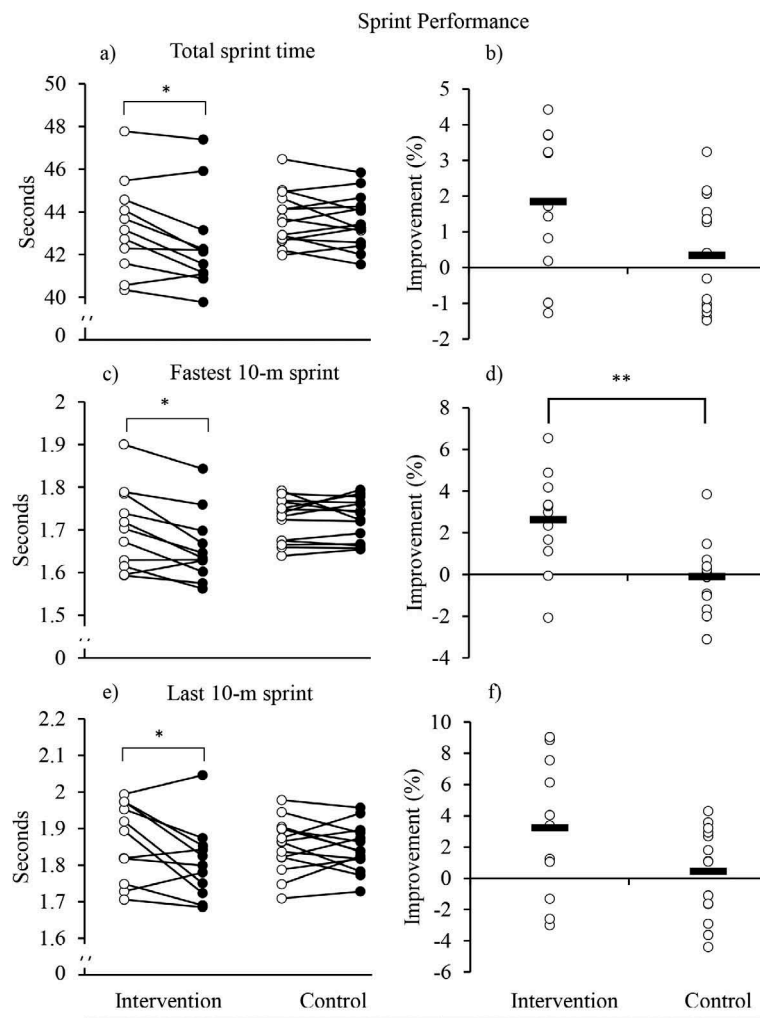


Figure 2. Sprint performance in the intervention- and control group. (a,c,e): Circles and black lines show paired measurements from baseline (white circles) to follow-up (black circles) in absolute sprint time (seconds) for each participant. (b,d,f): Circles show improvements in sprint performance in percent for each participant. Black bar shows mean change. * Denotes a statistically significant difference from baseline to follow-up ($p < 0.05$). ** Denotes a statistically significant between-group difference in mean change ($p = 0.005$).

Secondary outcomes measures

Analyses of between-group differences in mean changes showed a significant, and *very likely* (0/2/98), medium improvement in 10mST in favor of the intervention group (-0.045 s vs. 0.002 s; mean difference, -0.047 s, $p = 0.005$, $d = 0.64$), and a non-significant, but *likely* (2/11/87), medium improvement in L10mST in favor of the intervention group (-0.06 s vs. -0.008 s; mean difference, -0.052 s, $p = 0.094$, $d = 0.59$). In addition, the intervention group demonstrated pre-post improvements in TST, 10mST, and L10mST of 1.8%, 2.6%, and 3.2%, respectively ($p < 0.05$) (Figure 2(c-f)). The strength-related outcome measures showed a significant, and *almost certainly* (0/0/100), large between-group differences in mean change in ECC-P_{HS} (61.7 N vs. -9.3 N; mean adjusted difference, 62.3 N, $p = 0.006$, $d = 0.92$) and ECC-CAP_{HS} (769 N vs. -288 N, mean adjusted difference, 951 N, $p = 0.005$, $d = 0.95$) in favor of the intervention group. The intervention group demonstrated pre-post improvements in ECC-P_{HS} and ECC-CAP_{HS} of 19.2% and 17.4%, respectively ($p < 0.01$) (Figure 3(a-c)). No between-group difference in SFR_{HS} at follow-up (-4.5 N/rep vs. -3.5 N/rep; mean difference, -1.0 N/rep, $p = 0.363$) was observed. The

intervention group showed no pre-post change in SFR_{HS} (-3.3 N/rep vs. -4.5 N/rep; mean difference, -1.2 N/rep, $p = 0.449$) (Figure 3(c)). Likewise, a non-significant, and *unclear* (59/0/41), trivial between-group difference in mean change in AKE (2.2° vs. 1.4° ; mean difference, 0.8° , $p = 0.815$, $d = 0.07$) were observed. The intervention group showed no significant pre-post change in AKE (67.0° vs. 69.2° , $p = 0.478$). No significant pre-post changes were observed in the control group for any outcome measure ($p > 0.06$). A detailed overview of the results is provided in Table 3.

Harms

No harms during the execution of the NHE were observed.

Discussion

Sprint performance

The 10-week NHE strength training protocol *very likely* improved sprint acceleration performance (10mST) while *likely* improved repeated sprint performance (TST) and the last sprint

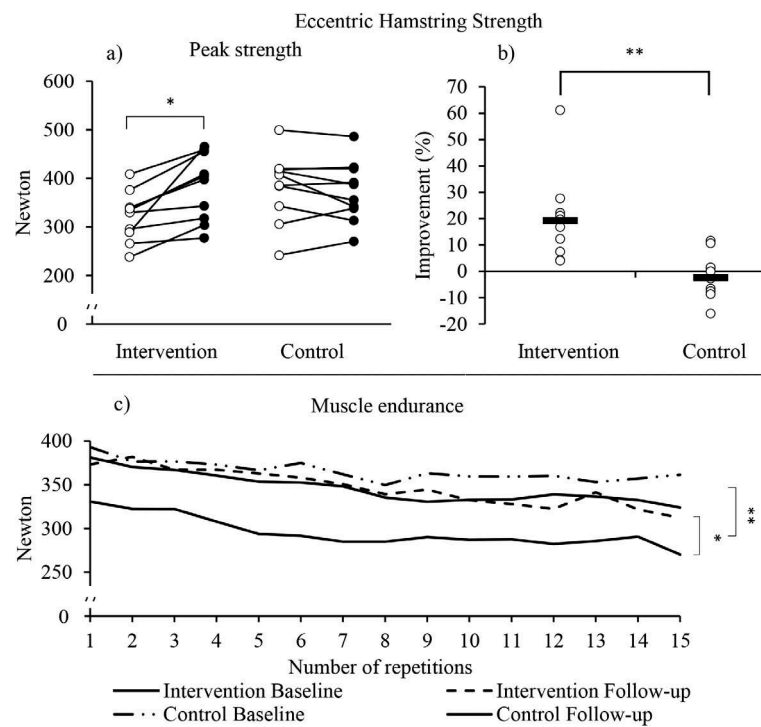


Figure 3. Eccentric hamstring strength and endurance in the intervention- and control group. (a) Circles and black lines show paired measurements from baseline (white circles) to follow-up (black circles) in absolute strength (Newton) for each participant. (b) Black bar shows mean change. (c) Lines show eccentric hamstring muscle strength during 15 repetitions of the Nordic Hamstring exercise in the intervention and control group at baseline and follow-up. * Denotes a statistically significant difference from baseline to follow-up ($p < 0.01$). ** Denotes a statistically significant between-group difference in mean change ($p < 0.01$).

performed (L10mST). The *likely-to-very likely* small-to-medium effect size improvements suggest a clinically relevant effect of both repeated- and 10 m sprint performance (Hopkins et al., 2009). As such, the present study confirms our preliminary findings on improved 10 m sprint performance using an identical NHE protocol (Krommes et al., 2017). Previous studies have reported either improved performance (Campos-Vazquez et al., 2015) or no performance enhancing effect (Buchheit et al., 2010) on repeated-sprint ability in football players following progressive heavy strength training using loaded squat exercise or explosive strength training using body weight loaded exercises, respectively. The above studies used a sprint distance of ≥ 30 m. The differences in sprint distance and strength training between protocols might explain the observed discrepancies between the present and previous study findings, since strength training seem to be more beneficial for short sprint distances ≤ 10 m (Rumpf, Lockie, Cronin, & Jalilvand, 2016). However, as the number of maximal short-distance sprints (≤ 10 m) have increased in top level football match play during the recent years (Barnes et al., 2014), maximal as well as repeated 10 m sprint capacity may be more crucial for match performance than more prolonged and much less frequent types of sprinting (Schimpchen et al., 2016). Thus, the improvement in TST may be considered a highly beneficial adaptation for the most intense periods of match play (Barnes et al., 2014).

The present improvements observed in secondary sprint-related outcome measures (10mST, L10mST) are in line with previous findings following strength training of the lower limb muscles (Rumpf et al., 2016). Reports exist of improved 5–30 m sprint performance in football players following specific

hamstring muscle strength training in isolation or in conjunction with other forms of training (Askling et al., 2003; Mendiguchia et al., 2015). Ten weeks of hamstring strength training using the YoYo™ flywheel device improved flying start 30m sprint performance by 2.4% (Askling et al., 2003). Additionally, a 7-week strength training program including eccentric hamstring exercises and plyometric training improved 5 m sprint performance by 1.6% (Mendiguchia et al., 2015). The present data indicate that the 10-week injury-prevention NHE protocol is equally efficacious in improving sprint performance compared to the above more extensive training programs. Conversely, the FIFA 11+ injury prevention program using a lower dose of the NHE compared to the present study did not report improved sprint performance (Haroy et al., 2017; Impellizzeri et al., 2013). Based on the mean between-group difference in 10mST and L10mST of -0.047 s and -0.052 s, respectively, an average player with a baseline sprint performance of approximately 1.7 s may likely gain a 25–30 cm lead during a maximal 10 m sprint. In football, such difference is considered clinically important, and may be crucial to reach a ball before the opponent player, block a shot, or score a goal (Haugen & Buchheit, 2016).

Potential mechanisms for improved sprint performance

Large and *almost certain* gains in the secondary strength-related outcome measures, ECC-P_{H5} and ECC-CAP_{H5} (17% and 19%), were observed following NHE training whereas no change was observed for SFR_{H5}. The present increase in ECC-P_{H5} confirms previous gains of 11% and 25% reported

in football players following a 10-week NHE intervention 1–3 (Mjolsnes et al., 2004) or 2 times per week (Bourne et al., 2017), respectively. The gain in ECC-CAP_{HS} suggests an increased ability to uphold high absolute eccentric force production during fatiguing contractions (Timmins et al., 2014). In combination with the gain in ECC-P_{HS}, these observations may suggest that the associated improvements in sprint performance result from an increased horizontal ground reaction force production (Morin et al., 2015). Debate on hamstring mechanics during high-speed running exists (Van Hooren & Bosch, 2017), however, available data suggest that the hamstring myofibers undergo a forceful stretch-shortening cycle (SSC) from the swing phase to the propulsive phase (ground contact) (Thelen, Chumanov, Best, Swanson, & Heiderscheit, 2005). The potential increase in horizontal force production could thus be due to enhanced SSC-efficiency (Cormie, McGuigan, & Newton, 2010). Speculatively, the present NHE training protocol may have involved neural adaptation mechanisms, such as increased spinal motor-neuron excitability, increased neural motor-drive, and increased rate of force development (Aagaard, 2003). Combined, this may have resulted in elevated levels of hamstring muscle activation during the terminal swing phase preparing the hamstring muscles to exert a large hip extensor moment upon foot contact with the ground thereby increasing horizontal force production during the propulsive ground contact phase, leading to improved sprint performance (Morin et al., 2015).

Potential for injury prevention

The specific role on low ECC-P_{HS} per se as a risk factor for HSIs is not fully clear (Timmins et al., 2016; van Dyk et al., 2017, 2016), however, the large increase in ECC-P_{HS} in the present study may suggest a decreased risk of HSIs (Timmins et al., 2016). In support of this notion, performing NHE training during the season 1–3 times per week effectively reduced the risk of HSIs by 60–85% in football players (Petersen et al., 2011; van der Horst, Smits, Petersen, Goedhart, & Backx, 2015). In addition to low ECC-P_{HS}, football players with low hamstring strength capacity may also be susceptible to HSIs (Schuermans et al., 2016). Fatigue-induced decrements in peak isometric and eccentric hamstring strength have been observed following football match play and repetitive maximal sprint running, respectively (Timmins et al., 2014; Wollin, Thorborg, & Pizzari, 2017). It may be speculated that an enhanced hamstring strength capacity may increase the margin to a potential critical low force threshold for increased HSI risk during fatiguing exercise (Timmins et al., 2016). Therefore, the increased ECC-CAP_{HS} in the present study might represent an important factor when considering the injury-preventive effect of the NHE (Petersen et al., 2011).

Clinical relevance

The improved sprint performance observed in the present study suggests that the NHE may satisfy health professionals focusing both on performance enhancement and injury-prevention of HSIs and therefore have important clinical implications for

health professionals working in sport environments (Gabbett & Whiteley, 2016). Previous claims have labelled the NHE as ineffective (Ekstrand et al., 2013) or even “contraindicated” (Gambetta & Benton, 2006) in relation to injury prevention and performance, respectively. However, the available data suggest that the NHE can prevent HSIs and improve sprint performance.

Methodological considerations and limitations

Due to apparent between-group baseline differences in ECC-P_{HS} and ECC-CAP_{HS} highlighted during the review process, we decided on a minor deviation from the planned statistical analyses of these secondary outcome measures. The reported results of the ANCOVA are similar to the results computed by the original independent t-tests prior to the review process. The present study contains potential limitations: Although, the repeated-sprint test was designed to replicate modern football play and overcome previous limitations associated with repeated-sprint tests, we acknowledge that the validity of the primary outcome measure, TST, in relation to match performance remains unknown (Taylor et al., 2016). Secondly, the drop-out rate of 10 participants potentially reduced the statistical power thereby increasing the risk of statistical type-2 errors (TST, L10mST). However, we were able to show a clinically relevant *likely* (TST) and *very likely* (L10mST) improvement in both outcome measures. Thirdly, as the overall compliance rate to NHE training was 60% with only few players above 70% compliance we did not perform per-protocol analyses as planned. Thus, it can only be speculated if higher compliance rates ($\geq 70\%$) would result in further improvements in sprint performance and eccentric hamstring strength capacity, respectively. It should also be noted that training compliance was registered as the number of completed NHE sessions and not based on the actual training volume performed, and therefore only reflects an estimation of the NHE work performed. The low compliance rate was due to non-attendance, illnesses, and general time constraints reflecting daily practice in non-professional football clubs. Fourthly, we included amateur football players, thus it is unknown whether the present results can be extrapolated to players at higher or lower performance levels. However, comparable sprint performance and ECC-P_{HS} values have been reported in elite football players (Styles, Matthews, & Comfort, 2016; Timmins et al., 2016). Finally, individual lower-limb strength training was not registered during the intervention period, however, this potential confounder was expected to be equally distributed between the groups (Moher et al., 2010).

Conclusion

Ten weeks of supervised evidence-based injury-prevention eccentric strength training for the hamstring muscles using the NHE resulted in a non-significant ($p = 0.056$), but *likely*, small improvement in repeated-sprint performance (+1.8%) in male football players. Additionally, *likely-to-very likely* medium improvements were observed in 10 m sprint performance (+2.6–3.2%), and *almost certain* large improvements were observed in peak eccentric hamstring strength and strength capacity (+17–19%).

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Contribution

LI was the principal investigator and was involved in study design, recruitment, data collection, analyses, interpretation of results, and did the first draft of the manuscript. All co-authors were involved in study design, analyses, and interpretation of results. All co-authors provided valuable intellectual comments during the manuscript write up and approved the manuscript before submission.

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